to the assessment of NMC efficiency may be accepted.
Methane efficiency < 15%
Ethane efficiency > 98%
8.5.1 NMC methane efficiency
Methane calibration gas at a concentration typical of that to be measured is flowed through the FID analyser with and without the NMC bypassed. The methane efficiency, Em, is determined as:
Em = 1 CM (wCutter)
 CM (w/oCutter)
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4) AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
Annex 7, page 11
I:\MEPC\MEPC 83-17-Add.1.docx
Where:
CM (wCutter) HC concentration with CH 4 flowing through the NMC (ppmC1)
CM (w/oCutter) HC concentration with CH 4 bypassing NMC (ppmC1)
8.5.2 NMC ethane efficiency
Ethane calibration gas at a concentration typical of the expected non-methane hydrocarbon concentration to be measured is flowed through the FID analyser with and without the NMC bypassed. The ethane efficiency, Ee, is determined as:
Ee = 1 CE (wCutter)
 CE (w/oCutter)
Where:

---CHUNK---

SOURCE: MEPC.402(83).pdf
non-methane hydrocarbon concentration to be measured is flowed through the FID analyser with and without the NMC bypassed. The ethane efficiency, Ee, is determined as:
Ee = 1 CE (wCutter)
 CE (w/oCutter)
Where:
CE (wCutter) HC concentration with C 2H6 flowing through the NMC (ppmC1)
CE (w/oCutter) HC concentration with C 2H6 bypassing NMC (ppmC1)
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4) AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
Annex 7, page 12
I:\MEPC\MEPC 83-17-Add.1.docx
APPENDIX 2
ENGINE TEST REPORT AND TEST DATA CH4 AND/OR N2O CALCULATIONS
Engine
Manufacturer
Engine type
Emission abatement device
Engine family or Engine group
identification

Serial number
Rated power
Rated speed
Emissions test results:
Test cycle
CfCH4 weighted g/kg fuel
Cslip-CH4 weighted % (of the mass
of the methane
containing fuel
used by the
engine)
N2O (weighted) g/kg total fuel
Test identification
Date/time
Test site
Test number
Company
Date and place of report
Signature
Measurement equipment
Manufacturer Model Measurement
ranges
Calibration

---CHUNK---

SOURCE: MEPC.402(83).pdf
containing fuel
used by the
engine)
N2O (weighted) g/kg total fuel
Test identification
Date/time
Test site
Test number
Company
Date and place of report
Signature
Measurement equipment
Manufacturer Model Measurement
ranges
Calibration
Span gas
conc.
Deviation
of
calibration
Analyser
HC / CH 4
Analyser*
ppm %
N2O Analyser ppm %
CO Analyser ppm %
CO2 Analyser % %

O2 Analyser % %
Speed rpm %
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4)
AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
Annex 7, page 13
I:\MEPC\MEPC 83-17-Add.1.docx
Torque Nm %
Power, if
applicable
kW %
Fuel flow-liquid %
Fuel flow-gas %
Air flow %
Exhaust flow %
Temperatures
Charge air
coolant inlet
oC oC
Exhaust gas oC oC
Inlet air oC oC
Charge air oC oC
Fuel-liquid oC oC
Fuel-gas oC oC
Pressures
Exhaust gas kPa kPa
Charge air kPa kPa
Atmospheric kPa kPa
Vapour pressure
Intake air kPa %
Humidity
Intake air % %
* For FID+NMC
Make and model of NMC
Before measurement After measurement
NMC CH4 gas concentration ppmC ppmC
HC with CH4 through NMC ppmC ppmC

---CHUNK---

SOURCE: MEPC.402(83).pdf
Charge air kPa kPa
Atmospheric kPa kPa
Vapour pressure
Intake air kPa %
Humidity
Intake air % %
* For FID+NMC
Make and model of NMC
Before measurement After measurement
NMC CH4 gas concentration ppmC ppmC
HC with CH4 through NMC ppmC ppmC
HC with CH4 bypassing NMC ppmC ppmC
NMC methane efficiency Em

NMC C2H6 gas concentration ppmC ppmC
HC with C2H6 through NMC ppmC ppmC
HC with C2H6 bypassing NMC ppmC ppmC
NMC ethane efficiency Ee
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4)
AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
Annex 7, page 14
I:\MEPC\MEPC 83-17-Add.1.docx
Liquid fuel characteristics
Fuel type
Fuel properties: Fuel elemental analysis:
Density ISO 3675 kg/m3 Carbon % m/m
Viscosity ISO 3104 mm2/s Hydrogen % m/m
Water ISO 3733 % V/V Nitrogen % m/m
Lower heating
value/Hu
MJ/kg Oxygen % m/m
Sulphur % m/m
Gas fuel characteristics
Fuel type:
Fuel properties Fuel elemental analysis
Methane number EN16726:2015 Carbon % m/m